A Study on the Compatibility of Gerrymandering Metrics (WIP)

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Abstract
This paper examines how effective various metrics are for detecting Partisan Gerrymandering, looking at Efficiency Gap (EG), Declination, Partisan Bias, and Mean-Median Difference (MMD). In order to use several metrics for legal purposes, the metrics used must be compatible, which means they shouldn't contradict each other frequently. This paper examines how the metrics relate to each other in the context of the 2020 redistricting maps in the USA by running Pearson correlation and linear regression tests. Results showed that Partisan Bias and Declination, and Partisan Bias and MMD were not compatible at all ($p = 0.656, 0.198$), while the rest were somewhat compatible ($p < 0.001$). Of these, Declination and MMD, and Declination and EG were most compatible ($R^2 = 0.37, 0.48$). Given the size of the dataset, pairs that were compatible may not hold over large-scale datasets, but the results show conclusively that Partisan Bias is incompatible with Declination and MMD.

Keywords
Gerrymandering, Metrics, Efficiency Gap, Declination, Partisan Bias, Mean-Median Difference

Introduction
Gerrymandering is the process of redrawing and manipulating electoral district boundaries in a state to give one party a systematic, unfair advantage over the other [2]. Gerrymandering has been commonplace for over 2 centuries, and has been an accepted part of American politics for most of that time. There are various types of Gerrymandering, but the 2 most common types are Racial and Partisan Gerrymandering. Racial Gerrymandering manipulates boundaries based on the racial distribution of voters, while Partisan Gerrymandering manipulates boundaries based on partisan lean. Gerrymandering garnered significant attention from the public and news media in the 2010s and has since then been extremely controversial, starting with North Carolina.

North Carolina 2010s
In the 2012 US House elections in the state of North Carolina, Democrats won 50.60% of votes, more than half of the votes statewide, but ended up winning only 4 of the 13 seats, equivalent to 30.77% of seats. In the 2014 elections, Republicans won 55.39% (boosted by the fact that Democrats did not contest District 9), but received 76.92% of seats. In 2018, Republicans won 50.39% of the statewide vote to 48.39% for Democrats, almost equal. Yet still, the map gave Republicans a rock solid 10-3 majority. These disproportionate results raised the question of whether they were deliberate, and if so, whether the map should have been allowed. Results such as those in North Carolina can sometimes arise naturally due to political geography and spatial distribution of votes, but they can also occur due to Gerrymandering [3].

Covington vs North Carolina
In the North Carolina State Senate, similarly disproportionate results arose due to Republican Gerrymanders. Legal action was taken and there were disputes regarding whether the maps drawn were constitutional or unconstitutional. For example there was the lower Federal court case Covington vs North Carolina, which made its way to the Supreme Court of the US, where both courts ruled that North Carolina's map was an unconstitutional racial gerrymander and ordered for the map to be redrawn [4].

Common Cause vs Lewis

Republicans redrew North Carolina's Senate map in 2017, and in the Wake County Superior Court Case, Common Cause vs Lewis, the court ruled that the map drawn was an unconstitutional Partisan Gerrymander under North Carolina's Constitution [5]. In addition, the 2017 case Cooper vs Harris heard by the Supreme Court of the US agreed that the maps in North Carolina were redrawn with too much a focus on race [6].

Introduction to Wisconsin

In Wisconsin, similarly disproportionate results arose during the 2012 and 2014 US House elections. Republicans got 5 of the 8 seats both times, despite both parties both getting approximately 50% of the vote both times. In a lower Federal court of Wisconsin, the court ruled in the case Gill vs Whitford that Wisconsin's map was unconstitutional and ordered for the maps to be redrawn [7]. However, the ruling was appealed and went to the Supreme Court who ruled that the plaintiffs did not have sufficient standing to challenge the map.

Rucho vs Common Cause

Finally, in 2019, the Supreme Court ruled in the case Rucho vs Common Cause that, as Chief Justice John G. Roberts Jr. put it, "We [The majority in the Supreme Court] conclude that Partisan Gerrymandering claims present political questions beyond the reach of the federal courts" and that while partisan gerrymandering may be unjust, the Supreme Court could not address the issue. This means that the onus is on states to deal with the issue [8][9]. While North Carolina's Supreme Court ruled that Partisan Gerrymandering violated the North Carolina Constitution, there is no such ruling for Federal Law [9].

Alexander vs South Carolina NAACP

In 2024, in the case Alexander vs South Carolina NAACP, the state court ruled that South Carolina's 1st District was drawn with race as a predominant factor, but the 2nd and 5th districts were drawn with race as only a motivating factor and not the predominant one [10]. The court eventually ordered for the map to be redrawn due to District 1 [11]. The case was appealed to the Supreme Court, and the Supreme Court ruled that there was insufficient evidence to claim that the map was a Racial Gerrymander. The map makers (the defendants) claimed that affecting partisan leaning of the districts was their goal [12]. In line with the Rucho vs Common Cause ruling, the court reaffirmed that it would not have any jurisdiction over Partisan Gerrymandering [13].

Examples of Democratic Gerrymander

Gerrymandering is clearly a contentious issue and there have been examples of very strong Gerrymanders as seen above. While the above states were contentious due to alleged Gerrymandering to favour Republicans, there are also states such as Maryland and Illinois where Gerrymandering was used to favour Democrats. Any party can Gerrymander, and the issue is by no means limited to the scope of the background covered in this paper.

The Actual Process and Requirements of Redistricting (and Gerrymandering)

Per the Voting Rights Act, Supreme Court and other laws, there are a few main requirements for Redistricting.

Equal Population

Firstly, Districts must have roughly equal populations [14]. There are several ways to define this. Examples include taking the range of the populations and dividing by the mean population. Other ways include to take the Percentage Mean Standard Error and Percentage Standard Deviation.

Minority Protection
Secondly, maps cannot systematically dilute the voting power of minorities [14]. This includes racial minorities, which means Racial Gerrymandering is prohibited. There is no single widely accepted empirical definition of this.

Contiguity and Compactness
Districts must be contiguous and compact. All precincts in a district must be one connected shape. Not all states have a compactness or contiguity requirement, but many do [14]. The requirements are usually very vague and often boil down to suggesting an eyeballing test. There is no single metric accepted in a legal context for compactness. However, some of the most prominent ones are the Polsby score, Reock score, Moment of Inertia Metric, and the Know It When You See It metrics [15][16][17]. Compactness metrics are not the focus of this paper, but are very relevant in redistricting and detecting Gerrymandering.

Communities of Interest
Some states require that maps must preserve certain Communities of Interest. This can include towns, cities, counties or regions with a social, cultural or other common identity [14].

Packing and Cracking
When it comes to the actual process through which a group goes about drawing a Gerrymandered map to favour them, there are two known techniques known as Packing and Cracking.

Packing
Packing is the process of creating a few districts that very very heavily favour the opposing party. Packing is done so that as many of the opposing party's voters as possible can be forced into a few very safe districts, which means the rest of the state, including the majority of the population, leans in favour of the desired party. Through this process the opposing party's voters' influence is reduced in many districts [18].

Cracking
Cracking is the process of creating districts where the opposing party's voters are spread amongst several districts such that the opposing party has a large share of the vote, but not enough to outright win any of the districts [19]. In essence, this involves the creation of districts where the opposing party gets many votes but loses by a relatively narrow margin.

Metrics
Packing and Cracking are the two essential components to any Gerrymander. Detecting both effectively is the key to detecting a Gerrymandered map. There are several well-known metrics that are currently used as indicators for Gerrymandering that we outline below.

The Efficiency Gap

The Efficiency Gap (EG) measures the relative difference in *wasted votes* between two parties in an election and was introduced by Stephanopoulos and McGhee in 2014 (published in 2015) [20]. In a 2 way race in a district, the winning party needs to get 50% of the vote to win (ignoring the exceedingly rare case of ties). No greater percentage of votes is strictly needed as the margin of victory is irrelevant. Therefore, any surplus percentage of votes beyond 50% are considered wasted as they don't contribute towards a candidate's victory. Likewise for the losing side, their candidate has lost, so none of their votes contribute to their candidate's victory because their candidate had no victory. Therefore all of their votes are considered wasted. Summing the difference in wasted votes between the two parties across the state in all districts and dividing by the total number of votes cast yields the Efficiency Gap [20].
When a party's voters are packed into districts, they will have a very high vote share in districts they win, meaning high surpluses and many wasted votes while for the other party, very few votes contribute to a losing cause, so few votes are wasted. Likewise when they are cracked into districts that they lose, they will have a high vote share (close to but less than 50%) being wasted, while the other party has a small surplus as they will have a vote share only marginally greater than 50%. Therefore, the Efficiency Gap is centered around detecting packing and cracking [20].

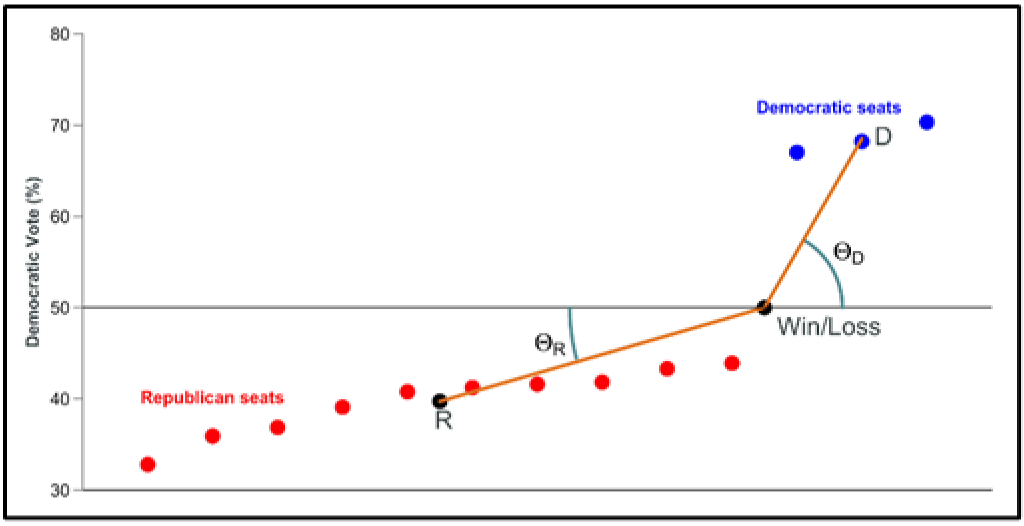
The Efficiency Gap has notably been successfully used to show Partisan Gerrymandering in legal contexts [21]. In the aforementioned case Gill vs Whitford, in the lower federal court case, the court sided with the plaintiffs and ruled that the map for the state legislature was an unconstitutional partisan Gerrymander. The plaintiffs used the Efficiency Gap to demonstrate that there was packing and cracking in the map and successfully showed that the map was a partisan Gerrymander. [22].

The main criticism of the Efficiency Gap is that it breaks down under circumstances of unequal voter turnout [28]. It is important to note that it is the major metric that has been used successfully in a Partisan Gerrymandering Case, most notable for its success in the lower court case of Gill vs Whitford.

Declination
Declination is another metric for gerrymandering developed by Warrington in 2018 [23]. It measures Gerrymandering by considering the geometric distribution of seats won and lost by a given party. Given an election, we have political parties P, Q and an n-vector, p, with the vote shares of party P. Without loss of generality, we let the components of p be ordered in monotonically increasing order. Let k be the highest index such that p_k < 1/2, meaning k' = N - k districts are won by party P. for i from 1 to k let A = {(i/N - 1/2N, p_i)} and for i from k + 1 to N let B = {(i/N - 1/2N, p_i}. The x coordinates range from -0.5 to 0.5 and are spaced equidistantly throughout the range, while the y coordinate represents the vote shares in the seats in increasing order. Let F = (k/2N, y_bar) and H = (k/N + k'/2N, z_bar) be the centers of masses of A and B respectively.
, let G = (k/N, 1/2). In a map that doesn't favour either side, G should lie on FH as it implies symmetry about the critical seat. If it lies above FH, this means that there is an advantage for party P.
The declination is the normalised difference in angles, $\frac{2}{\pi}\left|(\theta_P - \theta_Q)\right|$, where

$\theta_P = arctan(\frac{2\bar{z}-1}{k'/N})$ and $\theta_Q = arctan(\frac{1-2\bar{y}}{k/N})$

so that it lies between -1 and +1 [23]. Note that DRA uses the metric as an angle in degrees (-90 to +90)



Partisan Bias

The idea of Partisan Bias is that if the election is shifted uniformly to a tied result, no party should have an advantage.

Partisan Bias measures the deviation from equality if elections across a state were tied. Firstly, it shifts the election results across the state so that the results are 50-50 and applies that shift to each district. It then calculates the percentage of districts the parties win and how much it deviates from half.

The main criticism of Partisan Bias is that it relies on a *hypothetical* scenario which in less competitive states requires applying major shifts to results and assuming a uniform swing throughout a state, and is not grounded in reality (counterfactual). It is possible to have 2 maps with different seat shares in a real election, but with the same partisan bias.

Mean Median Difference

Mean Median Difference (MMD) takes all the districts and finds a party's vote share in the median district. It then compares its mean vote share (assuming equal population and equal turnout districts this is simply the statewide vote share) and calculates Median - Mean [23]. If the Mean Median Difference is positive for a party, the map is said to favour the party. Otherwise, the map is said to favour the other party.

Like Partisan Bias, Mean-Median Difference doesn't effectively respond to different seat shares [24]. While all metrics are not perfectly continuous, research has found that this problem is worse for Mean Median Difference and Partisan Bias [25].

While Federal courts may not be able to take measures to prevent Partisan Gerrymandering, individual states can. There is no single agreed upon way to quantify Partisan Gerrymandering in legal contexts. While the Efficiency Gap was used successfully by

Stephanopoulos in the lower court case of Gill vs Whitford, it has not been widely adopted. Stephanopoulos and McGhee, although they very strongly advocate for the Efficiency Gap, point out that it is not necessary to choose one single metric. There is no need to choose only one metric to use as an indicator, as all the metrics have their pros and cons. However, the main debate is over which metrics to choose and what should be the resulting threshold value. For example, in Gill vs Whitford, it was proposed that maps with Efficiency Gaps of over 7% in favour of either party were significantly Gerrymandered.

We loosely define the term "compatibility" when referring to a set of metrics. If a set of metrics, with each of their threshold values, serve as an effective tool to detect Gerrymandering, the set is compatible. There are

Methodology

Research Aim
This paper aims to examine how good the subsets of the 4 metrics (Efficiency Gap, Declination, Partisan Bias, and Mean Median Difference) would be for use in a legal definition of Partisan Gerrymandering.

Data Collection
Dave's Redistricting Application (DRA) [1] is an online tool that allows the user to easily draw redistricting maps in States for Congressional and State Legislature Elections. The application also has maps of all the current official redistricting maps used.

Election Dataset
Given that individual results in a state vary from election to election, the value of the metrics provided will depend accordingly, so it is important to choose an election that is representative of a normal or average scenario. There isn't a perfect way to measure this, but taking an average of the 2016 and 2020 US Presidential Elections is a good starting point. In 2016 and 2020, Donald Trump and Joe Biden won in the Electoral College, each winning 306 Electoral Votes, and won the tipping point state (Wisconsin in both cases) by 0.77% and 0.63%. Therefore, taking an average of results from 2016 and 2020 would result in an almost perfectly tied election, hence giving a reasonable indicator of an average election. DRA provides datasets using the average of 2016 and 2020 results for which we can view the metrics. Hence, for each state's maps (Congressional, State House and Senate or Legislature), results for each of the 4 metrics were collected taking the mean of the 2016 and 2020 presidential election results.

Exclusion Criteria
We looked at each of the 4 metrics for each state's Congressional and State Legislature maps. States with 1 congressional district have no potential for Gerrymandering and so their congressional maps were excluded. States with 2 congressional districts will automatically have a Mean Median Difference of 0% because the median result is simply the Mean of the two districts which mean they are equal. In addition, the Partisan Bias will always be 0 because no party can win both districts if the statewide vote is shifted to 50-50, meaning each party wins 1 seat, and the Partisan Bias will be 0. As a result, these maps were excluded as well. In some cases, states may have an undefined Declination because almost

all or all of the districts vote for one party, and so these maps were excluded as well. All other maps were included in the data.

The maps will include some Democratic Gerrymanders, some Republican Gerrymanders, and other maps that are more fair. In addition, it will contain some states that are very lopsided in favour of Democrats, competitive states and some states that are very lopsided in favour of Republicans. This allows us to see how the metrics perform against each other over a wide variety of scenarios, as they need to be effective in almost all scenarios.

Data Analysis

Google Sheets and Datatab were used for the data analysis to run Pearson correlations test for positive correlation and Linear Regressions between the various metrics. These were used to determine whether each pair of metrics is correlated and whether each pair is strongly related.

Results

The key is that given a set of metrics used for a legal classifier, they must be all mutually compatible. Firstly, the metrics must generally agree with each other. In the vast majority of cases, if one of the metrics clearly says that a map is a Gerrymander in favour of a certain party, all or almost all of them should agree with this. However, the metrics should diverge in a few cases, because no one metric can catch Gerrymandering 100% of the time. In essence, it should be reasonably possible to achieve low values of all metrics, the metrics should agree with each other in general, and the metrics should be able to catch the mistakes of the other metrics.

Declination vs Efficiency Gap

The Pearson p-value for positive correlation was statistically significant to suggest positive correlation, at less than 0.001.

Table 1: Pearson Test for Positive Correlation between Efficiency Gap and Declination

	r	p
Efficiency Gap and Declination	0.69	<.001

The Linear Regression resulted in a moderately high R^2 of 0.48. Hence, the results suggest that the Partisan Bias and Efficiency Gap are positively correlated, and that the correlation is moderately strong.

Table 2: Regression Analysis of Impact of Efficiency Gap on Declination (N=111)

Source	B	SE B	t	p
Constant	8.43	1.16	7.3	<.001***
Efficiency Gap	1.61	0.16	10.02	<.001***
R^2		0.48		

Note: *p<0.10, **p<0.05, ***p<0.01; B = coefficient; SE B= standard error

Despite the promising results for Declination and Efficiency Gap, the story is slightly different for Partisan Bias and Efficiency Gap

Partisan Bias vs Efficiency Gap

The Pearson p-value for positive correlation was statistically significant to suggest positive correlation, at less than 0.001.

Table 3: Pearson Test for Positive Correlation between Efficiency Gap and Partisan Bias

	r	p
Efficiency Gap and Partisan Bias	0.36	<.001

The Linear Regression resulted in a low R^2 of 0.13. Hence, the results suggest that the Partisan Bias and Efficiency Gap are positively correlated, but that the correlation is very weak.

Table 4: Regression Analysis on Impact of Efficiency Gap on Partisan Bias (N=111)

Source	B	SE B	t	p
Constant	-0.12	0.52	-0.24	0.814
Efficiency Gap	0.3	0.07	4.08	<.001***
R^2		0.13		

Note: *p<0.10, **p<0.05, ***p<0.01; B = coefficient; SE B= standard error

With the mixed results above for Partisan Bias and Efficiency Gap, Mean Median Difference and Efficiency Gap yield almost identical results.

Mean Median Difference vs Efficiency Gap

The Pearson p-value for positive correlation was statistically significant to suggest correlation, at less than 0.001.

Table 5: Pearson Test for Positive Correlation between Efficiency Gap and Mean Median Difference

	r	p
Efficiency Gap and Mean-Median Difference	0.35	<.001

The Linear Regression resulted in a low R^2 of 0.13. Hence, the results suggest that the Partisan Bias and Efficiency Gap are positively correlated, but that the correlation is very weak.

Table 6: Regression Analysis on Impact of Efficiency Gap on Mean Median Difference (N=111)

Model	B	SE B	t	p
(Constant)	2.62	0.27	9.58	<.001***
Efficiency Gap	0.15	0.04	3.84	<.001***
R^2		0.12		

Note: *p<0.10, **p<0.05, ***p<0.01; B = coefficient; SE B= standard error

With results so far always yielding clear positive correlations, Partisan Bias and Declination give the first results of clear incompatibility.

Partisan Bias vs Declination

The Pearson p-value for positive correlation was 0.656, which is not statistically significant and suggests that the two metrics are definitely not positively correlated.

Table 7: Pearson Test for Positive Correlation between Declination and Partisan Bias

	r	p
Declination and Partisan Bias	-0.04	.656

The Linear Regression resulted in an R value of 0.04 and a very low R^2 value very close to 0. With the coefficient of Declination being very close to 0 (in fact, it was slightly negative). This supports the results of the Pearson test, suggesting that the metrics are definitely not positively correlated.

Table 8: Regression Analysis on Impact of Declination on Partisan Bias (N=111)

Source	B	Standard error	t	p
Constant	0.55	0.66	0.83	0.406
Declination	-0.01	0.03	-0.4	0.689
R^2		0.00		

Note: *p<0.10, **p<0.05, ***p<0.01; B = coefficient; SE B= standard error

Meanwhile, Mean Median Difference and Declination give a stark contrast.

Mean Median Difference vs Declination

The Pearson p-value for positive correlation was statistically significant to suggest correlation, at less than 0.001.

Table 9: Pearson Test for Positive Correlation between Mean Median Difference and Declination

	r	p
Declination and Mean-Median Difference	0.61	<.001

The Linear Regression resulted in a moderately high R^2 value of 0.37. Along with the results of the Pearson test, this suggests that Mean Median Difference and Declination are definitely postively correlated, and that the strength of the correlation is moderate, like with Declination and Efficiency Gap.

Table 10: Regression Analysis on Impact of Declination on Mean Median Difference (N=111)

Source	B	Standard error	t	p
Constant	1.63	0.27	5.97	<.001***
Declination	0.11	0.01	7.95	<.001***
R^2		0.37		

Note: *p<0.10, **p<0.05, ***p<0.01; B = coefficient; SE B= standard error

To finish the Results section, Mean Median Difference and Partisan Bias give us the 2nd pair that is definitely incompatible.

Mean Median Difference vs Partisan Bias

The Pearson p-value for positive correlation was 0.198, which is not statistically significant enough to claim that the two metrics are correlated.

Table 11: Pearson Test for Positive Correlation between Partisan Bias and Mean Median Difference

	r	p
Partisan Bias and Mean-Median Difference	0.08	.198

The Linear Regression resulted in a model with an R value of only 0.08 and a low R^2 value of 0.01, supporting the Pearson test, suggesting that there is little to no correlation between Mean Median Difference and Partisan Bias.

Table 12: Regression Analysis on Impact of Partisan Bias on Mean Median Difference (N=111)

Source	B	Standard error	t	p
Constant	2.86	0.28	10.14	<.001***
Partisan Bias	0.04	0.05	0.85	0.397
R^2		0.01		

Note: *p<0.10, **p<0.05, ***p<0.01; B = coefficient; SE B= standard error

Discussion

In this section, we look at the results for each pair of metrics and comment on how the results compare to prior literature and whether each pair of metrics is compatible.

Efficiency Gap vs Declination

The comparison between Efficiency Gap and Declination gives a statistically significant p-value of less than 0.001 for the Pearson Correlation Test and an R^2 of 0.48. This suggests that the two metrics generally are in agreement with each other, as shown by the large cluster of points near the origin, but do occasionally disagree, meaning one suggests that there is a Gerrymander, while the other does not. This is an agreement with Stephanopolous and McGhee, who note that Efficiency Gap and Declination often behave similarly given that they both are based on the idea of detecting packing and cracking [24]. The divergence between the two metrics is also in agreement with Campisi et al., who point out that sometimes Declination cannot detect Gerrymandering when the Efficiency Gap can [25]. However, the data also supports the converse, as there are data points with high absolute value Efficiency Gap and low absolute value of Declination as well as vice versa. Overall, the results suggest that the two metrics are reasonably compatible for use as a legal requirement.

Mean Median Difference vs Partisan Bias

Looking at the comparison between the other two metrics, Mean Median Difference and Partisan Bias, the positive correlation test gives a p-value of 0.198, which suggests a rejection of the alternate hypothesis at a significance level of 0.05. In addition, the low R^2 value of 0.01 strongly supports the idea that the two are not correlated. This is quite surprising, given that, as Stephanopoulos and McGhee point out, the two metrics are quite similar based on their definitions. Partisan Bias is the difference between seats and 50% at 50% votes share, while Mean Median Difference is the difference between votes and 50% at 50% seat share [24]. However, the two metrics clearly aren't strongly correlated enough to make it reasonably easy to create a map where both of them are low. In a competitive state, it is more achievable, because if both parties have around 50% of the vote, the map should be designed to give them 50% of seats and that solves the problem. However, as Stephanopoulos and McGhee note, the metrics are less effective in more polarised states [24].

Partisan Bias vs Declination

Comparing Partisan Bias and Declination, the Pearson correlation test yields a very high p-value of 0.607, providing statistically significant evidence to reject the alternate hypothesis at a 0.05 significance level. In addition, the coefficient of Declination is slightly negative and close to 0, suggesting that the two are not compatible metrics. This isn't a surprise based on the aforementioned observations of Stephanopoulos and McGhee. There has never been a strong mathematical connection between the two metrics in literature unless the vote share is close to 50%, where all metrics converge [24].

Partisan Bias vs Efficiency Gap

When comparing Partisan Bias to Efficiency Gap, the Pearson correlation test does yield a statistically significant p-value of less than 0.001, suggesting that the two are correlated. This is quite reasonable given that Stephanopoulos and McGhee developed the Efficiency Gap metric as an improvement upon Partisan Bias, which was previously the main metric [20]. As a result, there are some similarities (such as the fact that both consider the difference of something from 50%). The main difference is that Partisan Bias relies on a hypothetical scenario with uniform swing and it is a major one. This difference could explain why the R^2 value was very low at 0.13 , suggesting a relatively weaker correlation, which is not ideal if both metrics were to be used in a legal context.

Mean Median Difference vs Efficiency Gap

There is a similar story for Mean Median Difference and Efficiency Gap. The p-value for the Pearson positive correlation test was statistically significant, under 0.001. However, the R^2 value was low as well at 0.12. Unlike with Partisan Bias, there is no direct connection between Efficiency Gap and Mean Median Difference. However, given the similarities between the definition of Mean Median Difference and Partisan Bias, the results are not unexpected.

Mean Median Difference vs Declination

When comparing Mean Median Difference with Declination, the Pearson test gave a statistically significant p-value of less than 0.001, suggesting the two are correlated. In addition, the R^2 is relatively high at 0.37, comparable to between Efficiency Gap and Declination. This is quite surprising because the two metrics have very different mathematical definitions, and don't fall into the same group of metrics from [24]. Regardless, the results suggest that the two metrics would be quite compatible.

General understanding / Summary of Discussion:

Partisan Bias is entirely incompatible with Declination and Mean Median Difference, and is not ideal for use with Efficiency Gap, and as a result, it does not make sense to add it into the set of metrics for legal requirements. Efficiency Gap and Declination have a strong correlation, and so they are very compatible, as is, surprisingly, Declination and Mean Median Difference. Mean Median Difference and Efficiency Gap have a pretty weak correlation, so like with Partisan Bias, it is not an ideal pair. This leaves the options of Declination and Mean Median Difference or Declination and Efficiency Gap. From here, we cannot make very strong suggestions based on our results as both pairs were similarly compatible with each other, but prior literature would point towards choosing Efficiency Gap, as Declination and Efficiency Gap are generally viewed as more similar metrics and are both able to handle very partisan states [24].

Conclusion

More Literature Review on the Metrics

While this paper does not primarily compare the metrics based on their mathematical definitions, there have been many such reviews in literature already. Prior research has found that in general, most metrics (that aren't based on district shape), including the 4 metrics examined in this paper, work very well in competitive elections as they often converge to each other [24]. In this context, competitive is defined as an election with a less than 10 percentage point margin of victory. Over the past few decades in past elections, all 4 metrics seem to follow a Normal / Gaussian distribution centered around 0 (no advantage for either party) [25]. Given that normal distributions have much higher probability densities (exponentially higher) near the center, which in this case is at 0, it shows that most past maps are generally close to fair, which shows that it is reasonably easy to create maps that are close to fair according to the metrics. It also shows that neither party has benefitted more than the other over a long period of time, which is expected, and that the metrics don't skew in favour of either party. However, each metric has its own shortcomings.

Partisan Bias + Mean Median Difference

Interesting Findings

Prior literature suggested that Mean Median Difference and Partisan Bias would be strongly correlated, and so would Declination and Efficiency Gap. While the latter did seem to be the case for the 2020s maps, surprisingly there was almost no correlation between the former pair. In addition, Partisan Bias wasn't correlated with Declination and weakly correlated with Efficiency Gap, while Mean Median Difference was weakly correlated with Efficiency Gap. These findings aren't surprising, given that the mathematical definitions of the metrics in each pair are very different. However, Mean Median Difference and Declination were quite strongly correlated, despite being unrelated metrics. Prior literature doesn't suggest any relation between the metrics, and their mathematical definitions are very different.

Limitations

Given the relatively small dataset, the results in this paper may not be extremely close to the correlations between the metrics over several decades of maps. The two most compatible pairs of metrics, Declination and Efficiency Gap and Declination and Mean Median Difference, may not necessarily be compatible over a larger dataset. The results in this paper are not sufficient to suggest that these pairs would be effective in a legal context. The paper is more effective in showing that certain pairs of metrics are not compatible. The pairs that weren't statistically correlated (Partisan Bias and Mean Median Difference, Partisan Bias and Declination) are clearly not suitable to be used together, because the paper shows that the pairs break down for the 2020s maps. The various R^2 values for the remaining pairs may suggest that 2 of the pairs are more suitable than the others, but this is not conclusive.

Future Work

The work in this paper avoids the topic of compactness and shape based metrics, such as the Polsby-Popper and Reock Scores. This is not to say that such metrics are unimportant; in fact, they are used quite frequently as potential indicators of Gerrymandering. The problem of compatibility, however, remains, as if any of these are used in legal contexts, they shouldn't very frequently contradict any other metrics used. One example of a way would be to see if there exists an algorithm that can draw district boundaries that can consistently Gerrymander a state, while keeping a certain metric low or close to 0.

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